

SANTHOSH KUMAR

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PROFILE SUMMARY

AI Engineer with 4+ years of hands-on experience designing and deploying production-ready GenAI, Machine Learning, and intelligent document processing systems. Proven expertise in building retrieval-augmented generation (RAG) pipelines using ChromaDB and large language models, developing and evaluating ML classification models for predictive use cases, and integrating AI workflows with scalable backend services using Python and Flask. Experienced across insurance, finance, and healthcare domains, contributing to AI-driven document intelligence platforms such as DocSens360.ai. Known for strong problem-solving, debugging, and ownership, with a track record of improving system reliability and scalability using Redis for efficient background processing.

TECHNICAL SKILLS

Languages & Runtime: JavaScript, Node.js, Python

Frameworks: Flask

GenAI & Agentic Systems: RAG, AI Agents, Agentic AI Workflows, LangChain, LangGraph, CrewAI, MCP

Databases: MongoDB, PostgreSQL, ChromaDB, Redis

APIs: RESTful API design & development

Tools & Platforms: Git, GitHub, Visual Studio Code

PROJECTS

Intelligent Document Management System (IDMS) – Insurance

Tech Stack: Python, Flask, ChromaDB, Redis

- Contributed to the development of an AI-powered, web-based Intelligent Document Management System (IDMS) to replace a legacy desktop application (ImageRight) used for insurance document management.
- Designed solutions to handle high-volume insurance documents arriving daily through scanners and email, enabling automated ingestion and tracking.
- Implemented reliable asynchronous background processing to monitor and process incoming documents at scale without impacting application stability.
- Built AI-driven workflows to extract structured data from unstructured scanned insurance documents, enabling automatic document classification and organization.
- Automated the creation of hierarchical folder structures (Team → Department → Claim → Document Type) based on claim information and business rules, eliminating manual folder creation and document uploads.
- Integrated the platform with downstream insurance systems such as ICMS and DEA to store, retrieve, and synchronize claim-related data across systems.
- Collaborated with cross-functional teams to deliver end-to-end document lifecycle management, improving operational efficiency and reducing manual effort in claim processing.

DocSens360.ai – AI Document Intelligence Platform

Tech Stack: Python, Flask, ChromaDB, Redis

- Contributed to the development of DocSens360.ai, a configurable, domain-agnostic AI document intelligence platform designed to ingest, process, and extract insights from unstructured documents across multiple use cases.
- Built LLM-powered document understanding and querying capabilities using retrieval-augmented generation (RAG) with ChromaDB, enabling users to interact with document knowledge through chatbot-based workflows.
- Integrated GenAI workflows with backend services using Python and Flask, delivering scalable, production-ready APIs.
- Designed and implemented asynchronous background processing to handle high-volume document ingestion from multiple sources (Scanner, Email, Google Drive, OneDrive, and application uploads), ensuring system stability under heavy load.
- Used Redis and ChromaDB to efficiently manage structured metadata and unstructured document embeddings.
- Served as a code reviewer, maintaining code quality, improving maintainability, and enforcing best engineering practices.

Curology – Personalized Skincare Subscription Platform

Tech Stack: HTML, CSS, JavaScript, Node.js, MongoDB

- Worked as a Node.js developer on a personalized, prescription-based skincare subscription platform inspired by Curology, connecting users with licensed providers for custom skincare formulas.
- Built backend APIs to support the online consultation flow, including skin-concern questionnaires and photo uploads used to match users with a personalized treatment plan.
- Developed subscription and order management modules to handle recurring shipments, formula adjustments, and billing cycles.
- Implemented secure user authentication and profile management to safeguard sensitive health and consultation data.
- Designed MongoDB schemas to store user skin profiles, prescription formulas, and order history for efficient retrieval and updates.
- Built a messaging feature enabling users to communicate with their assigned provider for follow-ups and routine adjustments.
- Collaborated with the frontend team to deliver a seamless, responsive experience across the consultation, shop, and account management sections of the platform.

ShopKart – E-Commerce Web App

Tech Stack: React, Node.js, MongoDB

- Designed and developed a scalable e-commerce platform inspired by Flipkart, enabling seamless browsing, shopping, and transactions for users.
- Implemented core features such as product catalog, user authentication, shopping cart, and order tracking.
- Built using React on the frontend with Node.js and MongoDB on the backend for efficient, responsive performance.

INTERNSHIP EXPERIENCE

Backend Developer Intern — Agira Technologies

Chennai, Tamil Nadu | March 2022 – July 2022

- Learned and applied JavaScript and Node.js in a professional development environment.
- Created a demo e-commerce web application to practice full-stack development concepts.
- Worked as a backend developer on an internal office management system project.

EDUCATION

Master of Computer Application — University of Madras | 2022 – 2024 | 70%

Bachelor of Computer Science — University of Madras | 2018 – 2021 | 74%

Higher Secondary Certificate (HSC) — Don Bosco Matriculation Higher Secondary School | 2016 – 2018 | 70.75%

LANGUAGES

English, Tamil